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Sensorless Control of PMSM Based on State Observer and the Parameter Error Analysis

Liyi Li*, Guangjun Tan, Jiaxi Liu and Baoquan Kou
School of Electrical Engineering and Automation, Harbin Institute of Technology
Harbin, Heilongjiang, China 150080
Email: liliyi.hit@gmail.com, tangjtangj@126.com, liujiaxi_hit@163.com, koubq@hit.edu.cn
Telephone: (86) 13503615595, Fax: (86) 451–86403771

***Abstract*—As state observer has the advantages of simple structure and low noise, this paper proposes a sensorless control method based on the state observer for the permanent magnet synchronous motor (PMSM) and analyzes the impact of the parameter errors. The state observer is firstly designed for the PMSM, and the frequency pre-warping bilinear transform method is adopted simultaneously in the discretization of the state observer. Then, the observation errors are analyzed when the stator resistance error or inductance error exists. Finally, Simulation results are presented, and verify the effectiveness of the paper's design and analysis.**

I. Introduction

Permanent magnet synchronous motor (PMSM) has the advantages of high power density, high efficiency, and high torque to inertia ratio, so it has been widely used in the industrial applications nowadays. In order to achieve high-precision and high speed responsive control of PMSM, the realtime position information of the rotor is required by the controller. The rotor position can be detected by mechanical sensors mounted on the rotor shaft. However, utilizing of these sensors will increase the cost and reduce the reliability of the system, and may suffer some restrictions such as temperature, humidity, vibration, and so on. In order to overcome the shortcomings brought to the system by using sensors, a number of researchers have developed the research of PMSMs' control systems based on sensorless control technology. This technology is of important significance to improve the system reliability and environmental suitability, and has been becoming the research focus in the motor control field. So it is of certain theoretical and practical significance to develop the research of the sensorless control technology for PMSMs.

Till now, several kinds of sensorless control methods for motors have been worked out internationally either for any type of motors or for any specific control purpose. Regardless of which method to be applied, these methods can be mainly classified into the following two types: one is the set of high-frequency (HF) signal injection methods based on the impedance difference of the motor stator, and the other is the set of fundamental excitation methods based on the motor's back electromotive force (EMF).

The former methods are suitable for salient pole motors or non-salient pole motors with saliency effect caused by magnetic saturation [1]–[4]. These methods do not need to know about the motor's fundamental model and parameters. If the stator is injected with HF voltage signals, the rotor position can be estimated through signal processing by detecting the HF current signals in the stator. They are effective during zero- and low-speed operations, and deteriorating gradually as the speed increases. Furthermore, injection of the HF signals will certainly bring in HF noise, harmonic losses, and extra pulsation to the electromagnetic torque.

The latter methods estimate the rotor position by using the back EMF, so they are suitable during medium- and high-speed operations as the amplitude of the back EMF is larger. These methods need to know about the motor's parameters. The simplest implementation method of them is direct computing method [5], [6]. It has the fastest response speed, but brings in much high-frequency estimation noise due to the difference operation of the phase currents. Also, it is an open-loop method which will generate estimation errors if the parameter values of the motor can not be obtained accurately, so it is always adopted combined with other closed-loop methods. In order to avoid the difference of currents, another open-loop method called flux observer was employed in [7], [8]. Some approaches had been adopted to avoid drifting of the pure integrator, and they may slow down the response speed more or less [2], [9]–[13]. Extended Kalman filter (EKF) is one of the most widely used sensorless control methods with good dynamic performance and anti-jamming capability [14]–[19]. However, it needs a plenty of calculated amount, so it is difficult to be applied in practice. An improved version to EKF, Unscented Kalman filters (UKF), can also be seen in the latest researches [20]–[22]. Model reference adaptive control (MRAC) is another popular solution for sensorless control, which is based on Lyapunov stability or Popov hyperstability criterion [23]–[28]. N. Matsui et al first proposed a sensorless method by building an estimated coordinate system in the brushless DC (BLDC) motor control system [29], and then brought this method into the PMSM control system [30]. This method is based on the assumption of "electrical steady state", so it maybe diverges if the estimated error is large enough. Another category of frequently-used methods are state observers, including full and reduced order observers, and sliding mode observer whose control variables are discontinuous [31]–[33].

State observer has the advantages of simple structure and low noise, so it has been applied to the sensorless control systems of PMSMs [34]–[39]. This paper presents a complete

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realization process of the state observer for the sensorless control system of PMSM. First, the paper designs the state observer for the PMSM, and the frequency pre-warping bilinear transform method is adopted simultaneously in the discretization of the state observer to decrease the discretization error. Then, the paper analyzes the observation errors of the observer when the stator resistance error or inductance error exists respectively. Finally, Simulation results are presented, and verify the effectiveness of this paper's design.

II. Differential Equation Model of PMSM

The PMSM that this paper researches on is surface-mounted ($L_d = L_q = L$) and has only one pair of magnetic poles ($p = 1$), so the voltage equations of its stator three-phase windings in the static $\alpha\beta$ coordinate system can be written as

$$u_\alpha = L\frac{\mathrm{d}i_\alpha}{\mathrm{d}t} + Ri_\alpha + e_\alpha \tag{1}$$

$$u_\beta = L\frac{\mathrm{d}i_\beta}{\mathrm{d}t} + Ri_\beta + e_\beta \tag{2}$$

where

i_α, i_β	α- and β-axes stator currents;
u_α, u_β	α- and β-axes stator voltages;
e_α, e_β	α- and β-axes stator back EMFs;
L	stator inductance;
R	stator resistance.

e_α and e_β are given respectively as follows

$$e_\alpha = -\omega\psi_f \sin\theta \tag{3}$$

$$e_\beta = \omega\psi_f \cos\theta \tag{4}$$

where

ω	rotor (electrical) angular velocity;
θ	rotor (electrical) position angle;
ψ_f	amplitude of permanent magnet (PM) flux linkage.

(1) and (2) can also be expressed as follows by using vectors

$$L\frac{\mathrm{d}\boldsymbol{i}_\mathrm{s}}{\mathrm{d}t} = \boldsymbol{u}_\mathrm{s} - \boldsymbol{e}_\mathrm{s} - R\boldsymbol{i}_\mathrm{s} \tag{5}$$

where

$$\boldsymbol{i}_\mathrm{s} = \left(i_\alpha i_\beta\right)^\mathrm{T} \tag{6}$$

$$\boldsymbol{u}_\mathrm{s} = \left(u_\alpha u_\beta\right)^\mathrm{T} \tag{7}$$

$$\boldsymbol{e}_\mathrm{s} = \left(e_\alpha e_\beta\right)^\mathrm{T} \tag{8}$$

The mechanical equation of the motor can be expressed as

$$J\frac{\mathrm{d}\omega}{\mathrm{d}t} = T_e - F\omega - T_m \tag{9}$$

where

T_e	electromagnetic torque;
T_m	load torque;
J	rotor's moment of inertia;
F	friction coefficient.

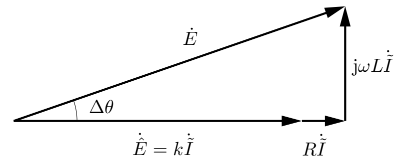


Fig. 1. Phasor diagram of the state observer

The electromagnetic torque T_e is calculated as

$$\begin{aligned} T_e &= \frac{3}{2\omega}\left(e_\alpha i_\alpha + e_\beta i_\beta\right) \\ &= \frac{3}{2}\psi_f\left(i_\beta\cos\theta - i_\alpha\sin\theta\right) \\ &= \frac{3}{2}\psi_f i_q \end{aligned} \tag{10}$$

where i_q is the q-axis active current.

III. Design of State Observer

As e_α and e_β contain the information of the rotor angular velocity and position angle, it is obvious to design the back EMF observer. Compared to the stator equation model (5), the state observer for the back EMF can be designed with the following form

$$\begin{cases} L\dfrac{\mathrm{d}\hat{\boldsymbol{i}}_\mathrm{s}}{\mathrm{d}t} = \boldsymbol{u}_\mathrm{s} - \hat{\boldsymbol{e}}_\mathrm{s} - R\hat{\boldsymbol{i}}_\mathrm{s} \\ \hat{\boldsymbol{e}}_\mathrm{s} = k\left(\hat{\boldsymbol{i}}_\mathrm{s} - \boldsymbol{i}_\mathrm{s}\right) = k\tilde{\boldsymbol{i}}_\mathrm{s} \end{cases} \tag{11}$$

where the variable with accent ˆ represents its estimated or measured value, and with accent ˜ represents its estimated or measured error; k is the observer gain. By subtracting (5) from (11), the error equation can be obtained as

$$L\frac{\mathrm{d}\tilde{\boldsymbol{i}}_\mathrm{s}}{\mathrm{d}t} = \boldsymbol{e}_\mathrm{s} - \hat{\boldsymbol{e}}_\mathrm{s} - R\tilde{\boldsymbol{i}}_\mathrm{s} = \boldsymbol{e}_\mathrm{s} - (k+R)\,\tilde{\boldsymbol{i}}_\mathrm{s} \tag{12}$$

From (12), we can obtain that the existence condition is

$$k > -R \tag{13}$$

The bigger value k is assigned, the higher convergence rate the observer reaches, and simultaneously, the more observation noise the observer brings in. To rewrite one equation of (12) in the phasor form as follows

$$\mathrm{j}\omega L\dot{\tilde{\boldsymbol{I}}} = \dot{\boldsymbol{E}} - \hat{\dot{\boldsymbol{E}}} - R\dot{\tilde{\boldsymbol{I}}} = \dot{\boldsymbol{E}} - (k+R)\,\dot{\tilde{\boldsymbol{I}}} \tag{14}$$

The phasor diagram representing (14) is shown in Fig. 1. From Fig. 1, we can come to a conclusion that the phase angle θ of the back EMF can be worked out by the sum of the phase angle $\hat{\theta}_u$ of the estimated back EMF and the compensatory angle $\Delta\theta$. $\hat{\theta}_u$ and $\Delta\theta$ are respectively derived as follows

$$\hat{\theta}_u = \mathrm{atan2}\left(-\hat{e}_\alpha, \hat{e}_\beta\right) \tag{15}$$

$$\Delta\theta = \arctan\frac{\omega L\tilde{I}_m}{k\tilde{I}_m + R\tilde{I}_m} = \arctan\frac{\omega L}{k+R} \tag{16}$$

From Fig. 1, according to the Pythagorean theorem we can get

$$\left(\hat{E}_m + R\tilde{I}_m\right)^2 + \left(\omega L\tilde{I}_m\right)^2 = {E_m}^2 \tag{17}$$

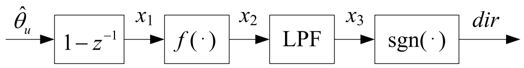


Fig. 2. Judgement of the rotation direction of the rotor

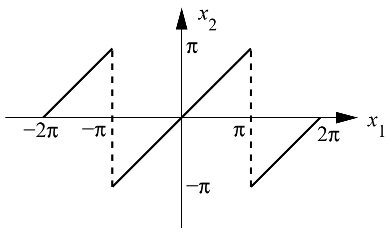


Fig. 3. Graph of function $f(\cdot)$

where

$$E_m = \omega\psi_f \tag{18}$$

$$\tilde{I}_m = \frac{\hat{E}_m}{k} \tag{19}$$

$$\hat{E}_m = \sqrt{\hat{e}_\alpha^2 + \hat{e}_\beta^2} \tag{20}$$

To substitute (18) and (19) to (17), we can derive the estimation formula for the rotor angular velocity ω

$$\hat{\omega} = \frac{(k+R)\,\hat{E}_m}{\sqrt{(k\psi_f)^2 - \left(L\hat{E}_m\right)^2}} \tag{21}$$

A low-pass filter (LPF) with the cut-off frequency of several times of switching frequency f_s, which has little impact on the dynamic response characteristic of the state observer, can be added for $\hat{\omega}$ to reduce the ripple caused by $\boldsymbol{u}_\mathrm{s}$ and $\boldsymbol{i}_\mathrm{s}$. As ω is unable to be measured directly, the ω value in (16) should be replaced by $\hat{\omega}$ derived from (21).

(15) and (21) can not judge the rotation direction of the rotor, and keep correct only when the rotor is in the forward rotation. This problem can be solved by checking the sign of the result of the difference operation of $\hat{\theta}_u$, which can also be added with an LPF to suppress the difference noise, as shown in Fig. 2. The difference operation is $1 - z^{-1}$ in Fig. 2, not $\left(1 - z^{-1}\right)/T_s$ like others. It can be seen obviously that the value range of x_1 is $(-2\pi, 2\pi)$. $f(\cdot)$ is designed to restrict the value range of x_2 to $[-\pi, \pi]$, and the graph of function $f(\cdot)$ is shown in Fig. 3. This transform is effective in almost all of the motor digital control systems (DCS) that the sampling frequency is more than twice as high as the motor's fundamental frequency. Furthermore, it can increase the dynamic response speed further. The cut-off frequency f_b of the LPF can be designed high because of low noise of state observer, so it has little effect in the start period of the motor and no effect after that period to the dynamic response characteristics. If $dir \in \{0, 1\}$, (15) and (21) are adopted; if $dir = -1$, (15) and (21) should be respectively changed into

$$\hat{\theta}_u = \mathrm{atan2}\left(-\hat{e}_\alpha, \hat{e}_\beta\right) + \pi \tag{22}$$

$$\hat{\omega} = -\frac{(k+R)\,\hat{E}_m}{\sqrt{(k\psi_f)^2 - \left(L\hat{E}_m\right)^2}} \tag{23}$$

The above judgement of rotation direction of the rotor maybe fails in the start period of the motor because of small output x_3 of LPF and relative high noise of the state observer. However, (21) and (23) have better dynamic and steady characteristics than calculating the angular velocity by simple discrete derivative and LPF processing of the estimated position angle. The block diagram of the state observer is shown in Fig. 4.

A. Discretization of the State Observer

For the continuous state observer (11), the discretization process should ensure that the digital observer after discretization has approximately the same dynamic and frequency response characteristics as (11) before discretization. Although a certain kind of discretization method may achieve completely or almost the same pulse response characteristic, it may not provide a preferable frequency response fidelity. For (11), the angular frequencies of all the input signals are the rotor's fundamental angular frequency ω, so it is only needed to ensure that the discretization method is able to achieve the same response characteristic at the angular frequency ω, not considering any other frequencies. This subsection will research on the discretization of the state observer.

As all the input signals of (11) have the same angular frequency ω, by taking α-axis for example we can assume that

$$\frac{\mathrm{d}\hat{\imath}_\alpha}{\mathrm{d}t} = u\left(\frac{t}{T}\right) = m\cos(\omega t + \phi) \tag{24}$$

where m and ϕ are defined constants. For continuous system, the solution of $\hat{\imath}_\alpha$ in the period $[(k-1)\,T, kT]$ is

$$\begin{aligned}\hat{\imath}_\alpha(k) &= \hat{\imath}_\alpha(k-1) + \int_{k-1}^{k} u\left(\frac{t}{T}\right)\mathrm{d}\left(\frac{t}{T}\right)\\ &= \hat{\imath}_\alpha(k-1) + \frac{2m}{\omega}\sin\frac{\omega T}{2}\cos\left(\frac{2k-1}{2}\omega T + \phi\right)\end{aligned} \tag{25}$$

For discrete system, the solution of $\hat{\imath}_\alpha$ in the period $[(k-1)\,T, kT]$ of the discretization method which has the same response characteristic at ω as the continuous system can be assumed to have the following form

$$\begin{aligned}\hat{\imath}_\alpha(k) &= \hat{\imath}_\alpha(k-1) + h\left[u(k-1) + u(k)\right]\\ &= \hat{\imath}_\alpha(k-1) + 2mh\cos\frac{\omega T}{2}\cos\left(\frac{2k-1}{2}\omega T + \phi\right)\end{aligned} \tag{26}$$

where h is the undetermined variable. By comparing (25) and (26), we can derive that

$$h = \frac{1}{\omega}\tan\frac{\omega T}{2} \tag{27}$$

By substituting (27) to (26), the discretization method can be expressed in the Z-domain as follows

$$\left(1 - z^{-1}\right)\hat{I}_\alpha(z) = \frac{1}{\omega}\tan\frac{\omega T}{2}\left(1 + z^{-1}\right)U(z) \tag{28}$$

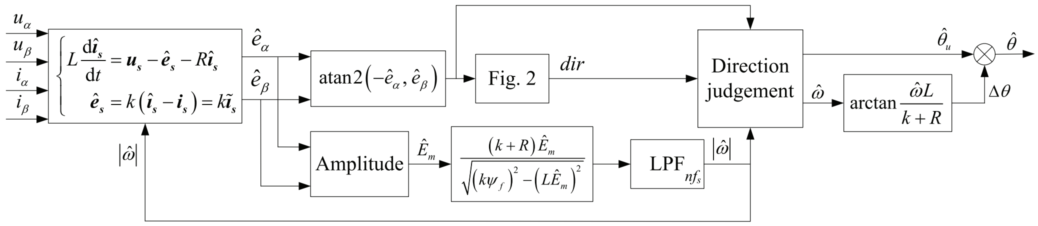


Fig. 4. Block diagram of the state observer

The expression of (24) in the S-domain is shown below

$$s\hat{I}_\alpha(s)=U(s) \tag{29}$$

The Z-transform formula which can ensure the same response characteristic at the angular frequency ω can be derived from (28) and (29)

$$s=\frac{\omega}{\tan\frac{\omega T}{2}}\frac{1-z^{-1}}{1+z^{-1}} \tag{30}$$

(30) is also known as frequency pre-warping bilinear transform method. It is an even function about the independent variable ω, so ω in (30) can be regarded as the absolute value of it, which is shown in Fig. 4 as the feedback channel from the signal $|\hat{\omega}|$ back to the state observer.

IV. Analysis on Parameter Errors to State Observer

State observer works well without parameter errors. However, the parameter errors exist due to measurement errors, and vary with the change of the working environment and operating condition. These errors will result in observation errors of the observer, and degrade the working performance of it. This section will analyze the influence of the error in the stator resistance or inductance to the state observer respectively (Here we assume $\omega \geq 0$, and the situation of $\omega < 0$ can be analyzed similarly and is not developed in the paper).

A. Analysis on Stator Resistance Error

When there is an error in the stator resistance, the model of the state observer is turned into

$$\begin{cases} L\dfrac{\mathrm{d}\hat{\boldsymbol{i}}_\mathrm{s}}{\mathrm{d}t}=\boldsymbol{u}_\mathrm{s}-\hat{\boldsymbol{e}}_\mathrm{s}-\hat{R}\hat{\boldsymbol{i}}_\mathrm{s} \\ \hat{\boldsymbol{e}}_\mathrm{s}=k(\hat{\boldsymbol{i}}_\mathrm{s}-\boldsymbol{i}_\mathrm{s})=k\tilde{\boldsymbol{i}}_\mathrm{s} \end{cases} \tag{31}$$

By subtracting (5) from (31), the error equation can be obtained as

$$L\frac{\mathrm{d}\tilde{\boldsymbol{i}}_\mathrm{s}}{\mathrm{d}t}=\boldsymbol{e}_\mathrm{s}-\hat{\boldsymbol{e}}_\mathrm{s}+R\boldsymbol{i}_\mathrm{s}-\hat{R}\hat{\boldsymbol{i}}_\mathrm{s}=\boldsymbol{e}_\mathrm{s}-\hat{\boldsymbol{e}}_\mathrm{s}-\tilde{R}\boldsymbol{i}_\mathrm{s}-\hat{R}\tilde{\boldsymbol{i}}_\mathrm{s} \tag{32}$$

To rewrite one equation of (32) in the phasor form as follows

$$\mathrm{j}\omega L\dot{\tilde{\boldsymbol{I}}}=\dot{\boldsymbol{E}}-\dot{\hat{\boldsymbol{E}}}-\tilde{R}\dot{\boldsymbol{I}}-\hat{R}\dot{\tilde{\boldsymbol{I}}}=\dot{\boldsymbol{E}}-\tilde{R}\dot{\boldsymbol{I}}_d-\tilde{R}\dot{\boldsymbol{I}}_q-\left(k+\hat{R}\right)\dot{\tilde{\boldsymbol{I}}} \tag{33}$$

The phasor diagram representing (33) is shown in Fig. 5 when $\tilde{R}>0, i_d>0, i_q>0$. The estimation formulas (21) and (16)

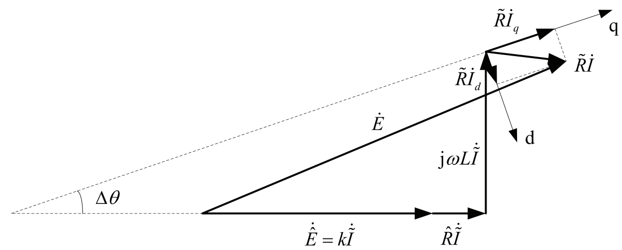


Fig. 5. Phasor diagram of the state observer ($\tilde{R}>0$)

should be changed into the following forms if the method without errors mentioned in III is still adopted

$$\hat{\omega}=\frac{\left(k+\hat{R}\right)\hat{E}_m}{\sqrt{(k\psi_f)^2-\left(L\hat{E}_m\right)^2}} \tag{34}$$

$$\Delta\theta=\arctan\frac{\hat{\omega}L}{k+\hat{R}} \tag{35}$$

As $\tilde{R}\neq 0$, the phase angle of $\dot{I}_q$ is no longer the same as $\dot{E}$. Assuming the phase difference that $\dot{I}_q$ leads $\dot{E}$ is $\Delta\phi$. According to geometry theory and based on (34) and (35), we can derive the approximate equations of $\hat{\omega}$ and $\Delta\phi$ about ω, i_d, i_q, and $\tilde{R}$ by Taylor expansion respectively as follows

$$\hat{\omega}\approx\omega-\frac{\tilde{R}i_q\left(k^2+2kR+L^2\omega^2+R^2\right)}{\psi_f(k+R)^2} \tag{36}$$

$$\Delta\phi\approx\frac{\tilde{R}}{\psi_f}\left(\frac{i_d}{\omega}-\frac{Li_q}{k+R}\right) \tag{37}$$

B. Analysis on Stator Inductance Error

When there is an error in the stator inductance, the model of the state observer is turned into

$$\begin{cases} \hat{L}\dfrac{\mathrm{d}\hat{\boldsymbol{i}}}{\mathrm{d}t}=\boldsymbol{u}-\hat{\boldsymbol{e}}-R\hat{\boldsymbol{i}} \\ \hat{\boldsymbol{e}}=k(\hat{\boldsymbol{i}}-\boldsymbol{i})=k\tilde{\boldsymbol{i}} \end{cases} \tag{38}$$

By subtracting (5) from (38), the error equation can be obtained as

$$\boldsymbol{e}-\hat{\boldsymbol{e}}-R\tilde{\boldsymbol{i}}=\hat{L}\frac{\mathrm{d}\hat{\boldsymbol{i}}}{\mathrm{d}t}-L\frac{\mathrm{d}\boldsymbol{i}}{\mathrm{d}t}=\tilde{L}\frac{\mathrm{d}\boldsymbol{i}}{\mathrm{d}t}+\hat{L}\frac{\mathrm{d}\tilde{\boldsymbol{i}}}{\mathrm{d}t} \tag{39}$$

To rewrite one equation of (39) in the phasor form as follows

$$\dot{\boldsymbol{E}}-\dot{\hat{\boldsymbol{E}}}-R\dot{\tilde{\boldsymbol{I}}}=\dot{\boldsymbol{E}}-(k+R)\dot{\tilde{\boldsymbol{I}}}=\mathrm{j}\omega\tilde{L}\dot{\boldsymbol{I}}_d+\mathrm{j}\omega\tilde{L}\dot{\boldsymbol{I}}_q+\mathrm{j}\omega\hat{L}\dot{\tilde{\boldsymbol{I}}} \tag{40}$$

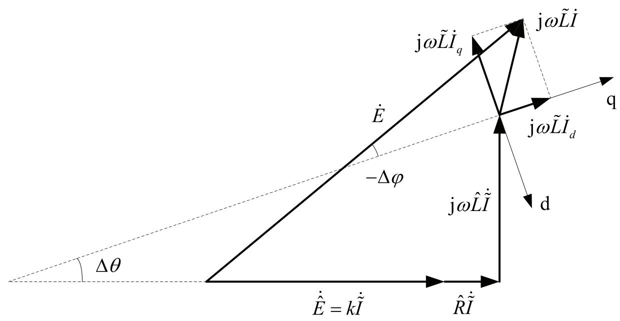


Fig. 6. Phasor diagram of the state observer ($\tilde{L} > 0$)

The phasor diagram representing (40) is shown in Fig. 6 when $\tilde{L} > 0, i_d > 0, i_q > 0$. The estimation formulas (21) and (16) should be changed into the following forms if the method without errors mentioned in III is still adopted

$$\hat{\omega} = \frac{(k+R)\,\hat{E}_m}{\sqrt{(k\psi_f)^2 - \left(\hat{L}\hat{E}_m\right)^2}} \tag{41}$$

$$\Delta\theta = \arctan\frac{\hat{\omega}\hat{L}}{k+R} \tag{42}$$

As $\tilde{L} \neq 0$, the phase angle of $\dot{I}_q$ is no longer the same as $\dot{E}$. Still with the assumption that the phase difference that $\dot{I}_q$ leads $\dot{E}$ is $\Delta\phi$, the included angle between $\dot{E}$ and q-axis is $-\Delta\phi$, and the included angle between $\dot{E}$ and d-aixs in Fig. 6 is $\frac{\pi}{2} + \Delta\phi$. According to geometry theory and based on (41) and (42), we can derive the approximate equations of $\hat{\omega}$ and $\Delta\phi$ about ω, I_m, and $\tilde{L}$ by Taylor expansion respectively as follows

$$\hat{\omega} \approx \omega - \frac{\tilde{L}\omega i_d\left(k^2 + 2kR + L^2\omega^2 + R^2\right)}{\psi_f\,(k+R)^2} \tag{43}$$

$$\Delta\phi \approx -\frac{\tilde{L}}{\psi_f}\left(i_q + \frac{\omega L i_d}{k+R}\right) \tag{44}$$

From (36), (37), (43), and (44), we can come to a conclusion that when there are errors in the stator resistance and resistance simultaneously, the estimated values of $\hat{\omega}$ and $\Delta\phi$ can be approximately expressed by the following equations

$$\hat{\omega} \approx \omega - \frac{(k+R)^2 + (\omega L)^2}{\psi_f\,(k+R)^2}\left(\tilde{R}i_q + \omega\tilde{L}i_d\right) \tag{45}$$

$$\Delta\phi \approx \frac{\tilde{R}i_d - \omega\tilde{L}i_q}{\omega\psi_f} - \frac{L\left(\tilde{R}i_q + \omega\tilde{L}i_d\right)}{(k+R)\,\psi_f} \tag{46}$$

If the $i_d = 0$ control scheme is adopted, (45) and (46) can be simplified to be

$$\hat{\omega} \approx \omega - \frac{\tilde{R}i_q}{\psi_f\,(k+R)^2}\left[(k+R)^2 + (\omega L)^2\right] \tag{47}$$

$$\Delta\phi \approx -\frac{\tilde{L}i_q}{\psi_f} - \frac{\tilde{R}Li_q}{(k+R)\,\psi_f} \tag{48}$$

TABLE I
PARAMETERS OF THE PMSM AND PWM INVERTER

Symbol	Meaning	Value
U_{dc}	DC bus voltage	300V
f_s	switching frequency	18kHz
p	number of pole pairs	1
ψ_f	permanent magnet flux linkage	0.0198Vs
J	rotor's moment of inertia	3.9385×10^{-5}kgm^2
T_m^*	rated load torque	0.1573Nm
F	friction coefficient	2.933×10^{-7}Nms
R_s	stator resistance	0.372Ω
L_s	stator inductance	0.937mH

TABLE II
PARAMETERS OF THE STATE OBSERVER AND CONTROLLERS

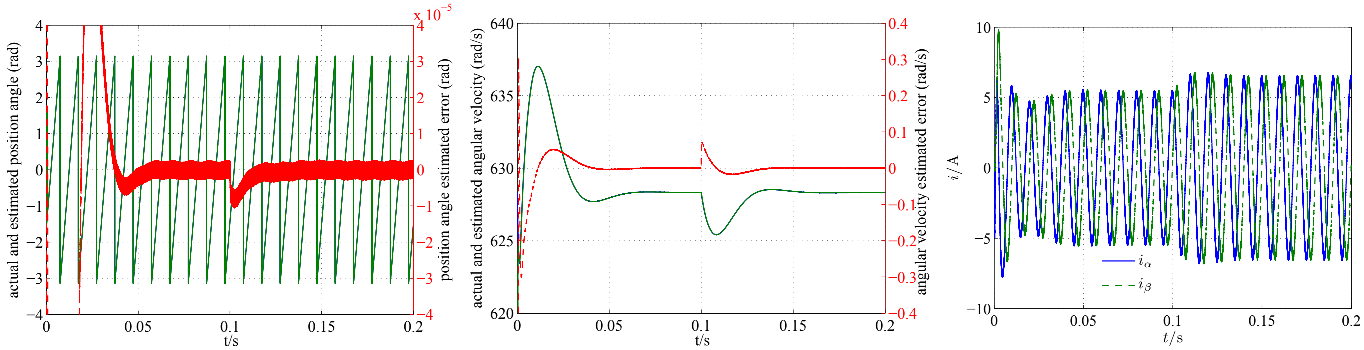
Symbol	Meaning	Value
k	observer gain	10V/A
f_b	cut-off frequency of LPF	10kHz
dq-axes current PI controllers		
K_{Ip}	proportional gain	1.2V/A
K_{Ii}	integral gain	2500V/(As)
velocity PI controller		
$K_{\omega p}$	proportional gain	0.23As
$K_{\omega i}$	integral gain	24A
$I_{\max}$	upper saturation limit	7.8125A
$I_{\min}$	lower saturation limit	-7.8125A

V. SIMULATION RESULTS

This section will present the simulation results of the proposed state observer and its parameter error analysis. The parameters of the PMSM and pulse width modulation (PWM) inverter are shown in Table I, and the parameters of the state observer and the current and velocity proportional-integral (PI) controllers designed for the PMSM are shown in Table II. The values of the PI gains are selected to ensure that the response time of i_d and i_q is about 1ms, and that of ω is about 10ms. The saturation limit outputs of the velocity controller are set to restrict the amplitude of i_q, which can avoid the occurrence of overcurrent.

A. Validition of the State Observer

First, the initial angular velocity ω_0 and the command angular velocity ω^* are both set as 200πrad/s to neglect the transient process of the rotor, and the load torque T_m is set to turn from T_m^* into 1.2 times of T_m^* at 0.1s to verify the anti-disturbance capability of the state observer. The simulation waveforms are shown in Fig. 7. According to Fig. 7(a) and Fig. 7(b), the rotation direction of the rotor is judged incorrectly before 0.39ms, and correctly after that time, so it can correctly judge the rotation direction after several sampling periods. It can be seen that the estimated position angle and angular velocity curves (green lines) coincide with the actual curves (blue lines), so the estimated values can track upon the actual values well both in the steady and transient state. From Fig.



(a) position angle and estimated error

(b) angular velocity and estimated error

(c) $\alpha\beta$-axes currents

Fig. 7. Waveforms when the motor runs at 200πrad/s

7(a), the position angle estimated error (red line) is zero in the steady state, -1.14×10^{-3}rad (settling time is 0.06s) in the start period, and -8×10^{-6}rad (recovery time is 0.02s) when the load torque varies at 0.1s respectively. From Fig. 7(b), the angular velocity estimated error (red dashed line) is zero in the steady state, 0.3rad/s (settling time is 0.07s) in the start period, and 0.071rad (recovery time is 0.06s) at 0.1s respectively. It can also be seen from Fig. 7(b) that the angular velocity overshoot is 1.39% in the start period, and the transient drop is 0.46% at 0.1s. Through analysis of the simulation results, we can see that the state observer works very well in any operation state, and has an excellent anti-disturbance capability to the load torque variation.

In order to verify the effectiveness of the judgement of rotation direction, the initial angular velocity ω_0 and the command angular velocity ω^* are then both set as -200πrad/s, and the simulation waveforms are shown in Fig. 8. The waveforms in Fig. 8 are almost opposite to Fig. 7, so the detailed analysis of the simulation waveforms of Fig. 8 is not developed and can refer to Fig. 7. The phase of i_α lags behind i_β in Fig. 8(c) which indicates the backward rotation, while the phase of i_α leads before i_β in Fig. 7(c) which indicates the forward rotation. From the simulation results of Fig. 8, we can see that the judgement of the rotation direction is still effective in the backward rotation.

B. Parameter Error Analysis when $i_d = 0$

In order to verify the conclusions of the parameter error analysis presented in this paper, the simulation results of the state observer are carried out here when the parameter errors exist. If there are 10% positive errors in the resistance and the inductance respectively, i.e. $\hat{R} = 1.1R$, $\hat{L} = 1.1L$, and the command angular velocity ω^* is set as 200πrad/s, the simulation waveforms are provided in Fig. 9 if the $i_d = 0$ control scheme is adopted. According to Fig. 9(a) and Fig. 9(b), it can be seen that the position angle estimated error $\Delta\phi$ (red line) and the angular velocity estimated error $\tilde{\omega} = \hat{\omega} - \omega$ (red line) are -0.026rad and -10.1rad/s respectively. According to (47) and (48), $\Delta\phi = -0.0263$rad, $\tilde{\omega} = -10.1027$rad/s. Therefore, (47) and (48) can correctly calculate the estimated errors when $i_d = 0$.

C. Parameter Error Analysis when $i_d \neq 0$

If $id = 1$A, $\hat{R} = 1.1R$, $\hat{L} = 1.1L$, and the command angular velocity ω^* is set as 200πrad/s, the simulation waveforms are provided in Fig. 10. According to Fig. 10(a) and Fig. 10(b), it can be seen that the position angle estimated error $\Delta\phi$ (red line) and the angular velocity estimated error $\tilde{\omega}$ (red line) are -0.024rad and -13.15rad/s respectively. According to (45) and (46), $\Delta\phi = -0.0236$rad, $\tilde{\omega} = -13.0857$rad/s. Therefore, (45) and (46) can correctly calculate the estimated errors when $i_d \neq 0$.

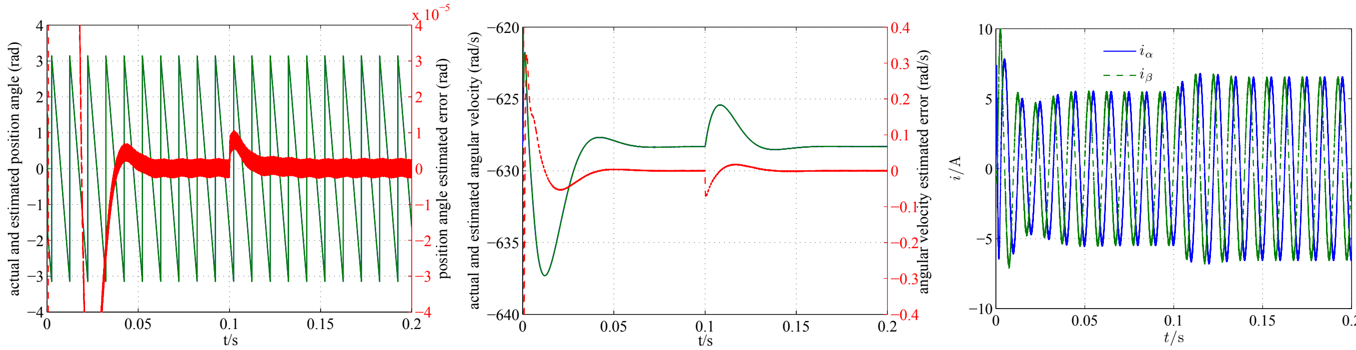
VI. Conclusion

This paper proposes a sensorless control method based on the state observer for the PMSM and analyzes the impact of the stator resistance error and inductance error. Simulation results show that the designed state observer can work well both in the steady and transient state with a good anti-disturbance capability. The simulation results also show that the position angle and the angular velocity estimated errors are in accordance with that of the error calculation equations when the stator resistance error or inductance error exists.

The future work will be divided into the following two directions: as the position angle and angular velocity estimated errors exist when the parameter errors exist, we will take some schemes to detect the parameter errors online so as to compensate these errors in the position angle and angular velocity in time; the second direction is to research on the parameter identification algorithm based on the state observer method to eliminate the parameter errors, and solve the problem of estimated errors at the beginning. Although several stator resistance identification methods have been adopted in the sensorless control of PMSM, they are not suitable to be applied in the state observer based sensorless control method, and the stator inductance identification methods except the complicated recursive least squares (RLS) method are not reported so far. So the second direction is still a difficult task.

References

[1] J.-H. Jang, S.-K. Sul, J.-I. Ha, K. Ide, and M. Sawamura, "Sensorless drive of surface-mounted permanent-magnet motor by high-frequency signal injection based on magnetic saliency," *Industry Applications, IEEE Transactions on*, vol. 39, no. 4, pp. 1031–1039, Jul.-Aug. 2003.

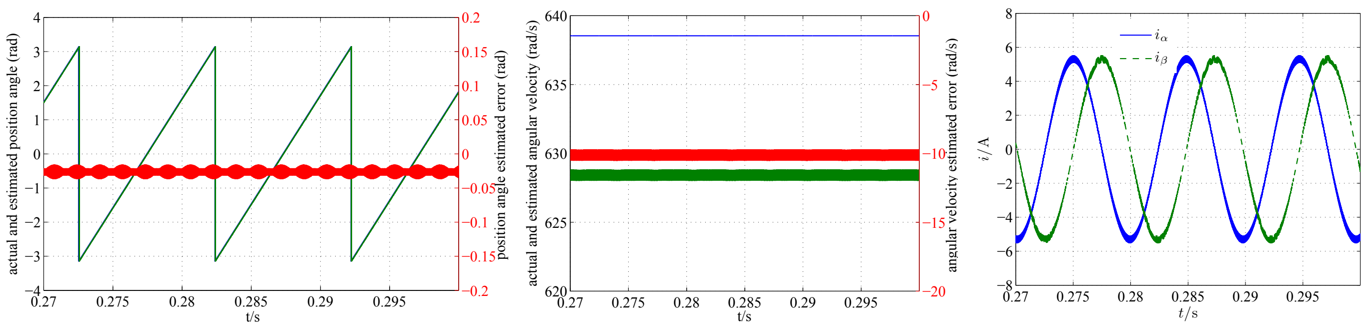


(a) position angle and estimated error

(b) angular velocity and estimated error

(c) $\alpha\beta$-axes currents

Fig. 8. Waveforms when the motor runs at -200πrad/s

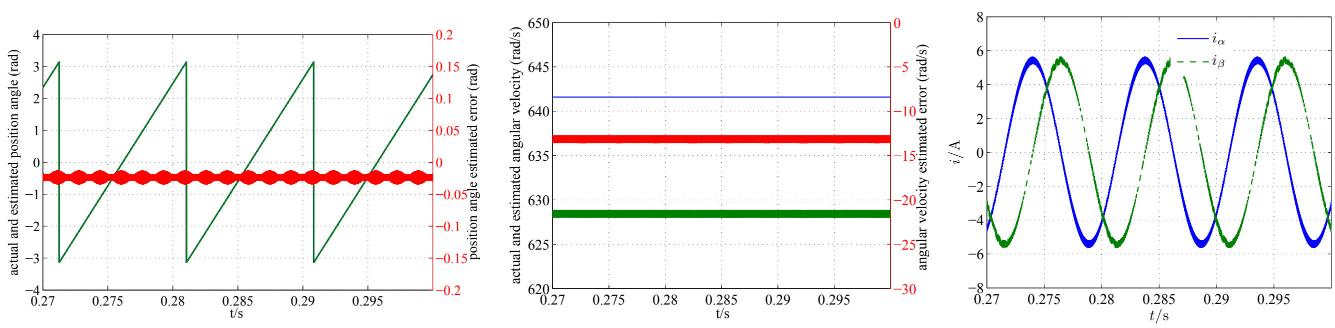


(a) position angle and estimated error

(b) angular velocity and estimated error

(c) $\alpha\beta$-axes currents

Fig. 9. Waveforms when parameter errors exist ($\hat{R} = 1.1R$, $\hat{L} = 1.1L$) and $i_d = 0$ control scheme is adopted



(a) position angle and estimated error

(b) angular velocity and estimated error

(c) $\alpha\beta$-axes currents

Fig. 10. Waveforms when parameter errors exist ($\hat{R} = 1.1R$, $\hat{L} = 1.1L$) and $i_d = 1$A

[2] C. Silva, G. M. Asher, and M. Sumner, “Hybrid rotor position observer for wide speed-range sensorless PM motor drives including zero speed,” *Industrial Electronics, IEEE Transactions on*, vol. 53, no. 2, pp. 373–378, Apr. 2006.

[3] J.-H. Jang, J.-I. Ha, M. Ohto, K. Ide, and S.-K. Sul, “Analysis of permanent-magnet machine for sensorless control based on high-frequency signal injection,” *Industry Applications, IEEE Transactions on*, vol. 40, no. 6, pp. 1595–1604, Nov.-Dec. 2004.

[4] J.-I. Ha, K. Ide, T. Sawa, and S.-K. Sul, “Sensorless rotor position estimation of an interior permanent-magnet motor from initial states,” *Industry Applications, IEEE Transactions on*, vol. 39, no. 3, pp. 761–767, May-Jun. 2003.

[5] M. A. Hoque and M. A. Rahman, “Speed and position sensorless permanent magnet synchronous motor drives,” in *Electrical and Computer Engineering, 1994. Conference Proceedings. 1994 Canadian Conference on*, vol. 2, Sept. 1994, pp. 689–692.

[6] M. Naidu and B. K. Bose, “Rotor position estimation scheme of a permanent magnet synchronous machine for high performance variable speed drive,” in *Industry Applications Society Annual Meeting, 1992., Conference Record of the 1992 IEEE*, vol. 1, Oct. 1992, pp. 48–53.

[7] M. C. Huang, A. J. Moses, and F. Anayi, “The comparison of sensorless estimation techniques for PMSM between extended Kalman filter and flux-linkage observer,” in *Applied Power Electronics Conference and Exposition, 2006. APEC ’06. Twenty-First Annual IEEE*, Mar. 2006, pp. 654–659.

[8] E. Urlep, J. Horvat, and K. Jezernik, “Pseudo sensorless control of PMSM,” in *Power Electronics and Motion Control Conference, 2006. EPE-PEMC 2006. 12th International*, Sept. 2006, pp. 1950–1955.

[9] G.-D. Andreescu, C. I. Pitic, F. Blaabjerg, and I. Boldea, “Combined flux observer with signal injection enhancement for wide speed range

sensorless direct torque control of IPMSM drives," *Energy Conversion, IEEE Transactions on*, vol. 23, no. 2, pp. 393–402, Jun. 2008.

[10] P. Guglielmi, M. Pastorelli, G. Pellegrino, and A. Vagati, "Position-sensorless control of permanent-magnet-assisted synchronous reluctance motor," *Industry Applications, IEEE Transactions on*, vol. 40, no. 2, pp. 615–622, Mar.-Apr. 2004.

[11] G. Foo and M. F. Rahman, "Sensorless direct torque and flux-controlled IPM synchronous motor drive at very low speed without signal injection," *Industrial Electronics, IEEE Transactions on*, vol. 57, no. 1, pp. 395–403, Jan. 2010.

[12] J. Yang, G. Yang, and T. Li, "Low cost sensorless vector control for PMSM compressors," in *Electrical Machines and Systems, 2008. ICEMS 2008. International Conference on*, Oct. 2008, pp. 1456–1459.

[13] S.-M. Jung, J.-S. Park, H.-W. Kim, and M.-J. Youn, "Improved rotor position estimation employing voltage distortion compensation for sensorless PMSM drives at low speed," in *Power Electronics, 2007. ICPE '07. 7th Internatonal Conference on*, Oct. 2007, pp. 811–816.

[14] S. Bolognani, L. Tubiana, and M. Zigliotto, "Extended Kalman filter tuning in sensorless PMSM drives," *Industry Applications, IEEE Transactions on*, vol. 39, no. 6, pp. 1741–1747, Nov.-Dec. 2003.

[15] M. Boussak, "Implementation and experimental investigation of sensorless speed control with initial rotor position estimation for interior permanent magnet synchronous motor drive," *Power Electronics, IEEE Transactions on*, vol. 20, no. 6, pp. 1413–1422, Nov. 2005.

[16] S. Bolognani, L. Tubiana, and M. Zigliotto, "EKF-based sensorless IPM synchronous motor drive for flux-weakening applications," *Industry Applications, IEEE Transactions on*, vol. 39, no. 3, pp. 768–775, May-Jun. 2003.

[17] A. Qiu, B. Wu, and H. Kojori, "Sensorless control of permanent magnet synchronous motor using extended Kalman filter," in *Electrical and Computer Engineering, 2004. Canadian Conference on*, vol. 3, May 2004, pp. 1557–1562.

[18] M. C. Huang, A. J. Moses, F. Anayi, and X. G. Yao, "Reduced-order linear Kalman filter (RLKF) theory in application of sensorless control for permanent magnet synchronous motor(PMSM)," in *Industrial Electronics and Applications, 2006 1ST IEEE Conference on*, May 2006, pp. 1–6.

[19] A. Akrad, M. Hilairet, and D. Diallo, "Design of a fault-tolerant controller based on observers for a PMSM drive," *Industrial Electronics, IEEE Transactions on*, vol. 58, no. 4, pp. 1416–1427, Apr. 2011.

[20] A. Cichowski, S. Bujacz, J. Nieznanski, and P. Szczepankowski, "Sensorless startup of super high speed permanent magnet motor," in *Industrial Electronics (ISIE), 2010 IEEE International Symposium on*, Jul. 2010, pp. 3101–3106.

[21] P. Borsjie, T. F. Chan, Y. K. Wong, and S. L. Ho, "A comparative study of Kalman filtering for sensorless control of a permanent-magnet synchronous motor drive," in *Electric Machines and Drives, 2005 IEEE International Conference on*, May 2005, pp. 815–822.

[22] T.-F. Chan, P. Borsje, and W. Wang, "Application of unscented Kalman filter to sensorless permanent-magnet synchronous motor drive," in *Electric Machines and Drives Conference, 2009. IEMDC '09. IEEE International*, May 2009, pp. 631–638.

[23] M. Rashed, P. F. A. MacConnell, A. F. Stronach, and P. Acarnley, "Sensorless indirect-rotor-field-orientation speed control of a permanent-magnet synchronous motor with stator-resistance estimation," *Industrial Electronics, IEEE Transactions on*, vol. 54, no. 3, pp. 1664–1675, Jun. 2007.

[24] T.-H. Liu, Y.-C. Lee, and Y.-H. Crang, "Adaptive controller design for a linear motor control system," *Aerospace and Electronic Systems, IEEE Transactions on*, vol. 40, no. 2, pp. 601–616, Apr. 2004.

[25] H. Jin and J. Lee, "An RMRAC current regulator for permanent-magnet synchronous motor based on statistical model interpretation," *Industrial Electronics, IEEE Transactions on*, vol. 56, no. 1, pp. 169–177, Jan. 2009.

[26] M. Tursini, R. Petrella, and F. Parasiliti, "Sensorless control of an IPM synchronous motor for city-scooter applications," in *Industry Applications Conference, 2003. 38th IAS Annual Meeting. Conference Record of the*, vol. 3, Oct. 2003, pp. 1472–1479.

[27] H.-W. Kim, M.-J. Youn, K.-Y. Cho, and H.-S. Kim, "Nonlinearity estimation and compensation of PWM VSI for PMSM under resistance and flux linkage uncertainty," *Control Systems Technology, IEEE Transactions on*, vol. 14, no. 4, pp. 589–601, Jul. 2006.

[28] H.-W. Kim, M.-J. Youn, and K.-Y. Cho, "New voltage distortion observer of PWM VSI for PMSM," *Industrial Electronics, IEEE Transactions on*, vol. 52, no. 4, pp. 1188–1192, Aug. 2005.

[29] N. Matsui and M. Shigyo, "Brushless DC motor control without position and speed sensors," *Industry Applications, IEEE Transactions on*, vol. 28, no. 1, pp. 120–127, Jan./Feb. 1992.

[30] R. Mizutani, T. Takeshita, and N. Matsui, "Current model-based sensorless drives of salient-pole PMSM at low speed and standstill," *Industry Applications, IEEE Transactions on*, vol. 34, no. 4, pp. 841–846, Jul./Aug. 1998.

[31] G. Foo and M. F. Rahman, "Sensorless sliding-mode MTPA control of an IPM synchronous motor drive using a sliding-mode observer and HF signal injection," *Industrial Electronics, IEEE Transactions on*, vol. 57, no. 4, pp. 1270–1278, Apr. 2010.

[32] J.-X. Xu, S. K. Panda, Y.-J. Pan, T. H. Lee, and B. H. Lam, "A modular control scheme for PMSM speed control with pulsating torque minimization," *Industrial Electronics, IEEE Transactions on*, vol. 51, no. 3, pp. 526–536, Jun. 2004.

[33] Y.-S. Han, J.-S. Choi, and Y.-S. Kim, "Sensorless PMSM drive with a sliding mode control based adaptive speed and stator resistance estimator," *Magnetics, IEEE Transactions on*, vol. 36, no. 5, pp. 3588–3591, Sept. 2000.

[34] M. Hinkkanen, T. Tuovinen, L. Harnefors, and J. Luomi, "A combined position and stator-resistance observer for salient PMSM drives: Design and stability analysis," *Power Electronics, IEEE Transactions on*, vol. 27, no. 2, pp. 601–609, Feb. 2012.

[35] A. Piippo, J. Salomaki, and J. Luomi, "Signal injection in sensorless PMSM drives equipped with inverter output filter," *Industry Applications, IEEE Transactions on*, vol. 44, no. 5, pp. 1614–1620, Sept./Oct. 2008.

[36] G. Bisheimer, M. O. Sonnaillon, C. H. De Angelo, J. A. Solsona, and G. O. Garcia, "Full speed range permanent magnet synchronous motor control without mechanical sensors," *Electric Power Applications, IET*, vol. 4, no. 1, pp. 35–44, Jan. 2010.

[37] G. Zhu, A. Kaddouri, L. A. Dessaint, and O. Akhrif, "A nonlinear state observer for the sensorless control of a permanent-magnet AC machine," *Industrial Electronics, IEEE Transactions on*, vol. 48, no. 6, pp. 1098–1108, Dec. 2001.

[38] M. Hasegawa and K. Matsui, "Position sensorless control for interior permanent magnet synchronous motor using adaptive flux observer with inductance identification," *Electric Power Applications, IET*, vol. 3, no. 3, pp. 209–217, May 2009.

[39] J. Solsona, M. I. Valla, and C. Muravchik, "A nonlinear reduced order observer for permanent magnet synchronous motors," *Industrial Electronics, IEEE Transactions on*, vol. 43, no. 4, pp. 492–497, Aug. 1996.